

2D simulation of fluid–structure interaction using finite element method

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ABSTRACT

This paper deals with pressure-based finite element analysis of fluid–structure systems considering the coupled fluid and structural dynamics. The present method uses two-dimensional fluid elements and structural line elements for the numerical simulation of the problem. The equations of motion of the fluid, considered inviscid and compressible, are expressed in terms of the pressure variable alone. The solution of the coupled system is accomplished by solving the two systems separately with the interaction effects at the fluid–solid interface enforced by an iterative scheme. Non-divergent pressure and displacement are obtained simultaneously through iterations. The Galerkin weighted residual method-based FE formulation and the iterative solution procedure are explained in detail followed by some numerical examples. Numerical results are compared with the existing solutions to validate the code for sloshing with fluid–structure coupling.

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1. Introduction

The transient response of liquid storage tanks due to external excitation can be strongly influenced by the interaction between the flexible containment structure and the contained fluid. The characteristics of the dynamic response of the flexible liquid storage tanks may be significantly different from that of the rigid liquid storage tanks. Hydrodynamic pressures are generated due to the fluid motion induced by the vibrating structures. These pressures modify the deformations, which in turn, modify the hydrodynamic pressures causing them. It has been observed that hydrodynamic pressure in a flexible container can be significantly higher than in the corresponding rigid container due to the coupling effects between the contained liquid and the elastic walls. The earlier theoretical studies on coupled slosh dynamics include both analytical and numerical treatments where circular cylindrical containers are studied most while the rectangular containers have received much less attention. The numerical treatments have mostly used the finite element technique for both liquid and structure motions. In most cases, the liquid is assumed inviscid and incompressible and the motion is irrotational. However, Muller [1] has shown that the compressibility of the liquid affects the frequency of the coupled system and the structure–compressible liquid system frequencies are lower than the structure–incompressible liquid system. In the reported studies, the structural displacements are almost invariably used to describe

the structural motion and the velocity potential function is found to be the most favored variable for representing the irrotational fluid motion. The hydrodynamic pressure is then required to be computed at each time step to determine the coupling forces acting on the structure. Use of the hydrodynamic pressure variable to represent the fluid motion has certain advantages in this context. First, in a pressure-based formulation, the compressibility of the liquid comes in a natural way and does not increase the computational difficulty and cost significantly. Secondly, the hydrodynamic pressure being the solution variable, the additional computational step of finding pressure, inherent in the potential-based formulations, is unnecessary. This can save considerable amount of computational time depending on the problem size and time integration technique employed.

The importance of the problem in several branches of engineering has attracted the attention of the researchers over the years and there exist a large number of theoretical and experimental studies on sloshing of contained liquid and the associated problems. The literature reports a variety of analytical and numerical techniques for formulating slosh models for different practical geometries. However, the most of the reported studies are concerned with rigid tanks. The structural flexibility and the free surface sloshing effects are not properly addressed in those studies. To the best knowledge of the present authors, a very few studies on analytical or numerical solutions of liquid sloshing problems in partially filled flexible containers with associated coupled interaction are reported in the open literature. Ibrahim [2], in his book, describes the fundamentals of liquid sloshing theory. The book describes systematically the basic theory and advanced analytical and experimental techniques in a

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self-contained and coherent format and deals with almost every aspect of liquid sloshing dynamics on space vehicles, storage tanks, road vehicle tanks and ships and elevated water towers under ground motion. An exhaustive literature survey is also included in the book. Morand and Ohayon [3] have presented two finite element methods for the computation of the variational modes of a system composed of an elastic tank partially filled with a compressible liquid. The authors have proposed a direct approach based on a three field mixed variational formulation and a variational modal interaction formulation allowing the use of the acoustic eigenmodes of the liquid in a rigid motionless enclosure and the hydroelastic modes of the enclosure. Haroun [4] has investigated the earthquake response of flexible cylindrical liquid storage tanks both numerically and experimentally. The structure and fluid domains are modeled using a finite element method and a Galerkin type method, respectively. The influence of static hoop stresses on wall vibration and the effect of the flexibility of the foundation are considered in the study. A number of researchers [1,5–7] have made use of the hydrodynamic pressure as the unknown variable in finite element discretization of the fluid domain. But the resulting equations in this case lead to unsymmetrical matrices and require a special purpose computer program [7]. Zienkiewicz et al. [5] represented the equations of fluid domain in terms of a displacement potential. The coupled equations of motion in this case become unsymmetrical, but the irrotationality condition on fluid motion is automatically satisfied. Liu and Ma [6] presented a coupled fluid–structure finite element method for the seismic analysis of liquid-filled systems considering the linearized free surface sloshing effect. Many researchers [8–10] have formulated the governing equations of fluid in terms of displacements. The advantage of the displacement-based formulation is that the fluid elements can easily be coupled to the structural elements using standard finite element assembly procedures. But especially for three-dimensional analysis the degrees of freedom for the fluid domain increase significantly. Moreover, the fluid displacements must satisfy the irrotationality condition, otherwise zero-frequency spurious modes may occur. Fenves et al. [11] have used both velocity and pressure variables for the governing equations of the fluid. However, with the increase in the number of unknown parameters in the fluid domain, the requirement of the computational time and storage increases rapidly. Thus the need of a large computer storage and expense of vast computer time usually make the analysis impractical. The solution of the coupled system may be accomplished by solving the two systems separately with the interaction effects enforced by iteration [12–14] or by a coupled solution [14]. The major advantage of the segregated method is that the coupled field problems may be tackled in a sequential manner. The analysis can be carried out for each field and updating the variables of the fields in the respective coupling terms accommodates interaction effect. Babu and Bhattacharyya [15] have developed a finite element numerical scheme to compute the free surface wave amplitude and hydrodynamic pressure in a thin walled container due to external excitation. Kim et al. [16] have presented an analytical study of liquid sloshing in three-dimensional rectangular elastic tanks. The authors have shown that the edge restraints on the walls of a three-dimensional rectangular vessel exert a significant influence on the dynamics of coupled fluid–structure interaction. However, the fundamental frequency of the coupled vibration mode rapidly approaches its two-dimensional value as the length to height ratio of a wall increases. This fact may justify the use of a two-dimensional model if adequate allowances are made. Particularly, for the dynamic analysis of a rectangular containment structure of the size typical for wet storage of nuclear spent-fuel assemblies, the two-dimensional model is expected to provide reasonable estimation of the coupled slosh characteristics. Koh et al. [17] have reported a variationally coupled BEM–FEM

formulation for the analysis of coupled slosh dynamics problem in two- and three-dimensional rectangular containers. The authors have successfully compared their computations with the conducted experiments. Bermudez et al. [18] have used finite element method to compute the sloshing modes in a rectangular rigid container with elastic baffle plates. The effect of the liquid motion is taken in account by means of an added mass formulation, discretized by standard piecewise linear tetrahedral finite elements.

Attempts have been made in the present study to analyze the coupled slosh dynamics in rectangular tanks with large length-to-height ratio. The two-dimensional model considers the cross section of the tank along the direction of the excitation and simulates the walls as cantilever beams. The motion of the contained liquid is represented through the small disturbance linearized wave equation presuming that the disturbance of the free surface is small in magnitude in comparison to the liquid depth and the wavelength so that the free surface conditions may be linearized. This has the inherent advantage that the free surface boundary is fixed in time, which simplifies the numerical solution procedure considerably. The assumption is quite justified when the exciting frequency is not very close to the natural sloshing frequency. The finite element technique is used to discretize both the structure and the fluid spatial domains. The finite element semi-discretized coupled equations are integrated in time using either a sequential predictor–multicorrector or a fully coupled algorithm. The finite element discretizations of the dynamical equations for the structure and fluid in the presence of the other and the two time integration techniques are discussed below. A few sample computations are included in this study.

2. Mathematical formulation

Sloshing analysis in elastic rectangular containers in two dimensions is carried out considering the sidewalls as cantilever beams. A typical liquid tank system is presented in Fig. 1. The bottom wall is treated as rigid. The hydrodynamic pressure on the walls arising due to the free surface oscillation causes the wall to deflect and move which in turn alters the free surface oscillation and the hydrodynamic forces on the wall. The two way interaction forces are shown in Fig. 2. In the present analysis the fluid is characterized by a single pressure variable and the coupling is achieved through a consideration of the interface forces. This method is widely used and has an advantage in the sense that in general a much smaller number of variables are involved to describe the fluid motion. The excess hydrodynamic pressure being the unknown variable, the interface

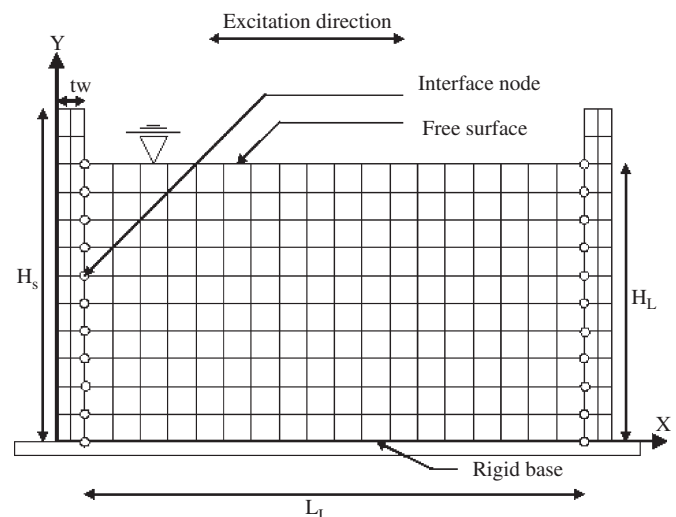


Fig. 1. Container and liquid domain with boundary and typical mesh.

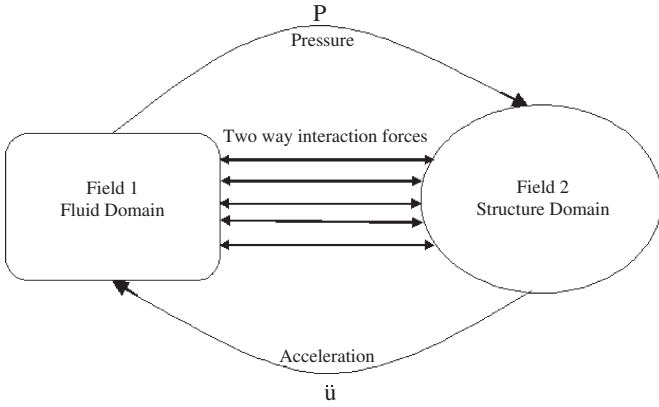


Fig. 2. Coupled field with interactive forces.

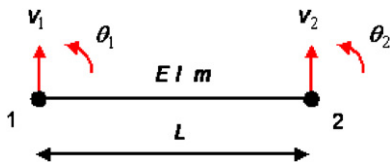


Fig. 3. A Bernoulli beam element.

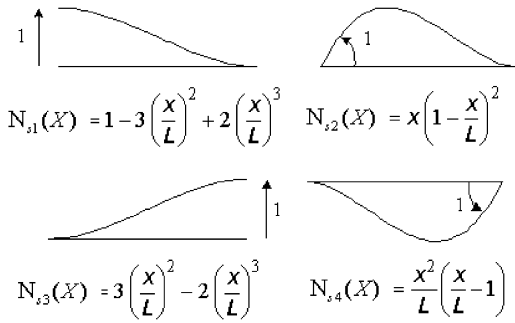


Fig. 4. Shape functions.

coupling forces at each time step can be computed directly, which can reduce the computational time significantly.

2.1. Structure domain

The container walls are discretized using Bernoulli beam elements with transverse and rotational deformations as shown in the Fig. 3. Stiffness and mass matrices for this element are represented by $[\mathbf{k}]$ and $[\mathbf{m}]$, respectively.

The mass per unit length of the structure element is $m = \rho A$, where ρ and A are the mass density and the cross sectional area of the beam element. The structural displacements and accelerations within an element are approximated using their nodal values as given by

$$v(x, t) = [\mathbf{N}_s] \{\mathbf{d}\} \quad \text{and} \quad \ddot{v}(x, t) = [\mathbf{N}_s] \{\ddot{\mathbf{d}}\}$$

where $\{\mathbf{d}\}$ is the vector of time dependent nodal displacements and

$$[\mathbf{N}_s] = [N_{s1}(x) N_{s2}(x) N_{s3}(x) N_{s4}(x)], \quad \{\mathbf{d}\} = \begin{Bmatrix} v_1(t) \\ \theta_1(t) \\ v_2(t) \\ \theta_2(t) \end{Bmatrix}$$

The interpolation functions (N_s) for the structural element are defined in Fig. 4 in an element Cartesian coordinate system.

The consistent element mass matrix for the beam element can then be written as

$$m_{ij} = \int_0^L m N_{si}(x) N_{sj}(x) dx$$

Assuming a linear elastic material with the stress–strain relation $\{\sigma\} = [\mathbf{E}]\{\epsilon\}$ and a strain–displacement relation $\{\epsilon\} = [\mathbf{B}]\{\mathbf{d}\}$, the elemental stiffness matrix can be obtained from the following relation:

$$k_{ij} = \int_0^L [\mathbf{B}]^T [\mathbf{E}] [\mathbf{B}] dx$$

On integration using the element shape functions, the elemental stiffness $[\mathbf{k}]$ and consistent mass matrices $[\mathbf{m}]$ are found to be as follows:

$$[\mathbf{m}] = \frac{mL}{420} \begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix}$$

$$[\mathbf{k}] = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

The finite element semi-discretized equation for the dynamics of the container structure can now be written in the familiar form given below [1,5,13,14]. No damping is considered in the motion of the structure

$$\mathbf{M}_s(\ddot{\mathbf{d}} + \ddot{\mathbf{u}}_g) + \mathbf{K}_s \mathbf{d} = \mathbf{F}_{\text{ext}} + \mathbf{Q}^T \mathbf{p} \quad (1)$$

with the globally assembled consistent mass matrix $\mathbf{M}_s$, stiffness matrix $\mathbf{K}_s$ and displacement vector $\mathbf{d}$. All externally applied loads are included in $\mathbf{F}_{\text{ext}}$. The fluid–structure coupling is represented by the term $\mathbf{Q}^T \mathbf{p}$, where $\mathbf{p}$ is the vector of the hydrodynamic pressure. The coupling matrix $\mathbf{Q}$ is given by

$$Q_{ij} = \int_{B_1} N_{fi}^T \hat{n}_s N_{sj} d\Gamma \quad (2)$$

where $\hat{n}_s$ gives the unit normal vector at the structure surface on the container–fluid interface. The shape functions for the structure and fluid domains are represented by $\mathbf{N}_s$ and $\mathbf{N}_f$, respectively. The base excitation or ground acceleration is denoted by $\ddot{\mathbf{u}}_g$.

2.2. Fluid domain

For sloshing of contained liquid, it has been observed that the effects of viscosity and compressibility of the fluid are usually very small, and most of the studies have successfully considered incompressible irrotational fluid motion with a high degree of accuracy [2–11,15–18]. Even though compressibility is found to have hardly any influence in the sloshing of a homogeneous fluid in a rigid container, it influences the sloshing response if the fluid is inhomogeneous and/or the container is elastic [1]. Based on these observations, the present finite element formulation considers an inviscid compressible homogeneous fluid and the governing equation, which is the well known wave equation, in terms of the excess pressure variable (p) is derived from the physical conservation laws. The equation is written as

$$\nabla^2 p(x, y, t) = \frac{1}{c^2} \ddot{p}(x, y, t) \quad \text{in } \Omega \quad (3)$$

where Ω is the fluid domain and c is the acoustic speed in the fluid. For two-dimensional motion in the (x, y) -plane with the excess

pressure $p(x, y, t)$, the equation can be explicitly written as

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} = \frac{\ddot{p}}{c^2} \quad \text{in } \Omega \quad (4)$$

The pressure formulation has certain advantages over the displacement or velocity potential-based formulations. Unlike the displacement formulation, the number of unknown in this formulation is only one per node, which results in considerable saving of computer storage and time. The saving will be more significant for large three-dimensional problems. In addition, the pressure field at the structure–fluid interface is directly obtained unlike displacement and potential formulations where the pressure has to be calculated from the velocity or displacements or their potential. This would be particularly advantageous in solving a fluid–structure interaction problem where pressure on the interface need to be computed at each time step. Besides these major advantages, the compressibility comes in a natural way in a pressure formulation and can be retained without incurring considerable additional efforts and costs.

The fluid boundary, in general, is composed of three types of boundaries. These are solid–liquid interface boundary, free surface boundary, non-reflecting or radiating type boundary. For liquid sloshing in a container the radiating type boundary is neglected. Fig. 1 shows the container configuration considered, the relevant boundaries, nomenclatures and definitions. The appropriate boundary conditions for these boundaries [5,13,19,20] are as follows:

1. *Solid–liquid interface boundary.* Continuity of normal displacement at the solid–liquid interface leads to the following relation for the linearized problem:

$$\frac{\partial p}{\partial n} = -\rho_f \ddot{u}_n \quad \text{on } B_{tw} \quad (5)$$

where B_{tw} stands for the tank wall. The interface boundary is $B_l = B_{tw}$. The normal acceleration of the interface is denoted by $\ddot{u}_n$.

2. *Free surface boundary.* The linearized free surface boundary condition is given by

$$p = \rho_f g \zeta \quad (6a)$$

$$\Rightarrow \frac{\partial p}{\partial n} = -\frac{\ddot{p}}{g} \quad \text{on } B_f \quad (6b)$$

3. *Bottom boundary.* Considering the bottom boundary to be rigid, at the bottom of the tank

$$\partial p / \partial n = 0 \quad \text{on } B_{tb} \quad (7)$$

The total boundary is $B = B_{tw} + B_f + B_{tb}$, as defined in Fig. 1.

The semi-discretized equation for the fluid motion in a container can be used for coupled slosh dynamics. The normal acceleration that appears in the equation through the fluid–structure interface boundary condition would now consist of the structural displacement (d) as well as the base excitation ($\ddot{u}_g$).

The integral form of the governing equation and the boundary conditions can be cast in to a weighted residual form as follows:

$$\int_{\Omega} \bar{N}^T \left[\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} - \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} \right] d\Omega - \int_{B_l} \bar{N}^T \left[\frac{\partial p}{\partial n} + \rho_f \ddot{u}_n \right] d\Gamma - \int_{B_f} \bar{N}^T \left[\frac{\partial p}{\partial n} + \frac{\ddot{p}}{g} \right] d\Gamma = 0 \quad (8)$$

where $\bar{N}^T$ is the weight function.

Using Green–Gauss theorem and introducing the finite element approximations the above equation reduces to the following

equation:

$$\int_{\Omega} \left[\frac{\partial N_i}{\partial x} \sum_1^N \frac{\partial N_j}{\partial x} \bar{p}_j(t) + \frac{\partial N_i}{\partial z} \sum_1^N \frac{\partial N_j}{\partial z} \bar{p}_j(t) \right] d\Omega + \int_{\Omega} \bar{N}_i^T \left[\frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right] \sum_1^N N_j \bar{p}_j(t) d\Omega + \int_{B_f} \bar{N}_i^T \left[\frac{1}{g} \frac{\partial^2}{\partial t^2} \right] \sum_1^N N_j \bar{p}_j(t) d\Gamma = \int_{B_l} \bar{N}_i^T [\rho_f \ddot{u}_n] d\Gamma$$

The above equation is rewritten as

$$\int_{\Omega} \left[\frac{\partial N_{fi}}{\partial x} \sum_1^{nf} \frac{\partial N_{fj}}{\partial x} \bar{p}_j(t) + \frac{\partial N_{fi}}{\partial y} \sum_1^{nf} \frac{\partial N_{fj}}{\partial y} \bar{p}_j(t) \right] d\Omega + \int_{\Omega} \bar{N}_{fi}^T \left[\frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right] \sum_1^{nf} N_{fj} \bar{p}_j(t) d\Omega + \int_{B_f} \bar{N}_{fi}^T \left[\frac{1}{g} \frac{\partial^2}{\partial t^2} \right] \sum_1^{nf} N_{fj} \bar{p}_j(t) d\Gamma = \int_{B_l} \bar{N}_{fi}^T [\rho_f \ddot{u}_n] d\Gamma \quad (9)$$

In the equation above, nf denotes the total number of fluid elements. The normal acceleration of the structure within an element at the solid–liquid interface can be approximated using the shape functions used for structural dynamics. Assuming the total number of structural elements to be ns , Eq. (9) can be written as

$$\int_{\Omega} \left[\frac{\partial N_{fi}}{\partial x} \sum_1^{nf} \frac{\partial N_{fj}}{\partial x} \bar{p}_j(t) + \frac{\partial N_{fi}}{\partial z} \sum_1^{nf} \frac{\partial N_{fj}}{\partial z} \bar{p}_j(t) \right] d\Omega + \int_{\Omega} \bar{N}_{fi}^T \left[\frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right] \sum_1^{nf} N_{fj} \bar{p}_j(t) d\Omega + \int_{B_f} \bar{N}_{fi}^T \left[\frac{1}{g} \frac{\partial^2}{\partial t^2} \right] \sum_1^{nf} N_{fj} \bar{p}_j(t) d\Gamma = \int_{B_l} \bar{N}_{fi}^T \rho_f \sum_{j=1}^{ns} N_{sj} (\ddot{d} + \ddot{u}_g) d\Gamma \quad (10)$$

The semi-discretized equations (10) for the fluid system can be written as

$$\mathbf{M}_f \ddot{\mathbf{p}} + \mathbf{K}_f \mathbf{p} = \mathbf{F}_f + \rho_f \mathbf{Q} (\ddot{\mathbf{d}} + \ddot{\mathbf{u}}_g) \quad (11)$$

where $\mathbf{M}_f$ and $\mathbf{K}_f$ are the assembled global mass and stiffness matrices for the fluid, respectively. The variable $\mathbf{d}$ represents the global structural displacements, $\mathbf{p}$ represents nodal pressures and $\mathbf{F}_f$ is the external load. The subscripts ‘s’ and ‘f’ refer to the solid and fluid domains, respectively. A superposed dot represents the time derivative. The coupling matrix $\mathbf{Q}$ transfers the acceleration of the structure to the fluid domain and the fluid pressure to the structure domain.

The coupled systems of equations are solved using two approaches. In the sequential or segregated approach, each system is solved separately with known solution of the other system. In the fully coupled or simultaneous approach, the two systems are solved simultaneously as a single system.

3. Time integration of the coupled field equations

3.1. Sequential predictor-multicorrector scheme

The governing second-order ordinary differential equation for either system at time step $(n+1)$ can be written as

$$\mathbf{M} \ddot{\mathbf{s}}_{n+1} + \mathbf{K} \mathbf{s}_{n+1} = \mathbf{F}_{n+1} \quad (12)$$

The subscripts are dropped as it may be used for either field. The force term augments applied force, specified boundary conditions

and interaction terms from the other field. In the predictor phase the field variables are expressed as

$$s_{n+1}^i = \tilde{s}_{n+1} \quad (13)$$

$$\dot{s}_{n+1}^i = \tilde{\dot{s}}_{n+1} \quad (14)$$

$$\ddot{s}_{n+1}^i = 0 \quad (15)$$

where i is the iteration count and

$$\tilde{s}_{n+1} = \dot{s}_n + \Delta t(1 - \gamma)\ddot{s}_n \quad (16)$$

$$\tilde{s}_{n+1} = s_n + \Delta t\dot{s}_n + \frac{1}{2}\Delta t^2(1 - 2\beta)\ddot{s}_n \quad (17)$$

Here γ and β are the Newmark's parameters and Δt is the time step. In the corrector solution phase the following equation is formed and solved:

$$K^* \Delta s_{n+1}^i = f_{n+1}^i \quad (18)$$

$$\text{where } K^* = M/\beta \Delta t^2 + K \quad (19)$$

$$\text{and } f_{n+1}^i = F_{n+1}^i - M\ddot{s}_{n+1}^i - Ks_{n+1}^i \quad (20)$$

Once the increment in the field variable is obtained, the field variable and its derivatives are updated as follows:

$$s_{n+1}^{i+1} = s_{n+1}^i + \Delta s_{n+1}^i \quad (21)$$

$$\dot{s}_{n+1}^{i+1} = (\dot{s}_{n+1}^i + \dot{s}_{n+1}^i/\beta\Delta t^2) \quad (22)$$

$$\ddot{s}_{n+1}^{i+1} = \ddot{s}_{n+1}^i + \gamma \Delta t \dot{s}_{n+1}^{i+1} \quad (23)$$

Finally a convergence check is made on the norm of the increment in the field variable compared to the norm of the total field variable as follows:

$$\text{Is } \frac{\|\Delta s^i\|}{\|s^{i+1}\|} \leq e \quad (e = \text{specified tolerance}) \quad (24)$$

If 'NO', $i \rightarrow i + 1$, and go to Eq. (18) for the next iteration.

If 'YES', $n \rightarrow n + 1$, and go to Eq. (12) for the next time step.

The stability criterion and hence the time step size of Newmark's integrator for a coupled problem depends on the mesh integrator, predictor formula and computational path. An illuminating analysis is included in Paul [14].

3.2. Fully coupled scheme (simultaneous solution)

Eqs. (1) and (11) can be combined to obtain the complete fluid–structure dynamic interaction equation as follows:

$$\begin{bmatrix} M_s & 0 \\ \rho_f Q & M_f \end{bmatrix} \begin{Bmatrix} \ddot{d} \\ \ddot{p} \end{Bmatrix} + \begin{bmatrix} K_s I & -Q^T \\ 0 & K_f \end{bmatrix} \begin{Bmatrix} d \\ p \end{Bmatrix} = \begin{Bmatrix} F_{\text{ext}} - M_s \ddot{u}_g \\ F_f - Q \ddot{u}_g \end{Bmatrix} \quad (25)$$

The solution of the above non-standard unsymmetric system needs specialized approaches. The difficulty can be circumvented through a rearrangement of Eq. (25) in a symmetric form.

The first row in Eq. (25), i.e., the coupled structural dynamical equation can be pre-multiplied by M_s^{-1} and rearranged to obtain the structural acceleration as

$$\ddot{d} = -M_s^{-1}K_s d + M_s^{-1}Q^T p + M_s^{-1}F_{\text{ext}} - M_s^{-1}M_s \ddot{u}_g \quad (26)$$

Substituting of the above equation in the second row of Eq. (25) results in

$$\rho_f Q(-M_s^{-1}K_s d + M_s^{-1}Q^T p + M_s^{-1}F_{\text{ext}} - M_s^{-1}M_s \ddot{u}_g) + M_f \ddot{p} + K_f p = F_f - Q \ddot{u}_g$$

or

$$-\rho_f Q M_s^{-1}K_s d + [K_f + \rho_f Q M_s^{-1}Q^T]p + M_f \ddot{p} + \rho_f Q(M_s^{-1}F_{\text{ext}} - M_s^{-1}M_s \ddot{u}_g) = F_f - Q \ddot{u}_g \quad (27)$$

The structural dynamical equation, on pre-multiplication with $K_s M_s^{-1}$, can be written in the following form:

$$K_s M_s^{-1}M_s \ddot{d} + K_s M_s^{-1}K_s d = K_s M_s^{-1}F_{\text{ext}} + K_s M_s^{-1}Q^T p - K_s M_s^{-1}M_s \ddot{u}_g \quad (28)$$

Eqs. (27) and (28) are coupled to be expressed in the following form:

$$\begin{bmatrix} K_s & 0 \\ 0 & M_f/\rho_f \end{bmatrix} \begin{Bmatrix} \ddot{d} \\ \ddot{p} \end{Bmatrix} + \begin{bmatrix} K_s M_s^{-1}K_s & K_s M_s^{-1}Q^T \\ Q M_s^{-1}K_s & K_f/\rho_f + Q M_s^{-1}Q^T \end{bmatrix} \begin{Bmatrix} d \\ p \end{Bmatrix} = \begin{Bmatrix} K_s M_s^{-1}(F_{\text{ext}} - M_s \ddot{u}_g) \\ F_f - Q \ddot{u}_g + Q M_s^{-1}(F_{\text{ext}} - M_s \ddot{u}_g) \end{Bmatrix} \quad (29)$$

The right-hand side vector in the above matrix equation is set to zero for free vibration analysis.

The implementation of the algorithm needs considerably larger core storage, since the matrices in the second term of Eq. (29) are full matrices unlike the original mass and stiffness matrices, which are banded. Hence, this approach requires a large amount of computation. To minimize the computation, in all the time dependent solutions presented in this study, the sequential approach is used. The fully coupled approach is used for free vibration studies only.

4. Results and discussion

The rectangular tank system considered for the study is 19.6 m wide and 12.3 m high with the depth of contained liquid (water) being 11.2 m. This, in fact, is the cross section of the 56 m × 19.6 m × 12.3 m rectangular tank considered in the analytical studies due to Kim et al. [16]. The tank walls are 1.2 m thick, which is usual for typical nuclear spent fuel storage tanks for the purpose of radioactive and thermal protection. The structural material has density of 2300 kg/m³ and modulus of elasticity of 2.0776×10^{10} Pa. The contained liquid is water with density of 1000 kg/m³. The coordinate system and some nomenclatures are defined in Fig. 1. Fig. 5 shows two mode shapes where the deformation of the sidewalls in a first cantilever mode is prominent. In Fig. 5a, both the walls move in the same direction (inward in this case) resulting in the deformed tank shape being symmetric and, hence, the mode is termed as a symmetric mode. One sidewall is the mirror image of the other wall with respect to the tank centerline. The deformation of the sidewalls in Fig. 5b is again in the first cantilever mode but the sidewalls now

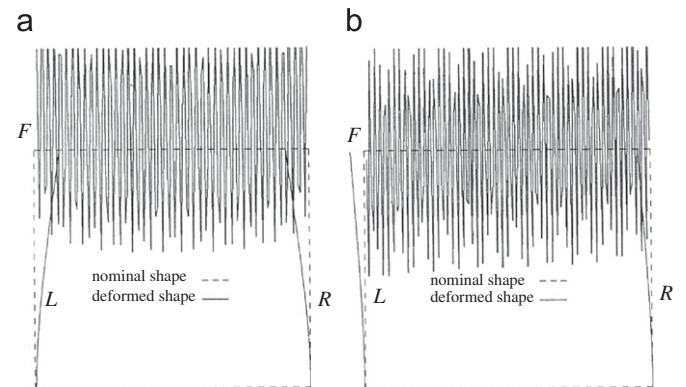


Fig. 5. Structural mode shapes. (a) First symmetric structural mode. (b) First anti-symmetric structural mode.

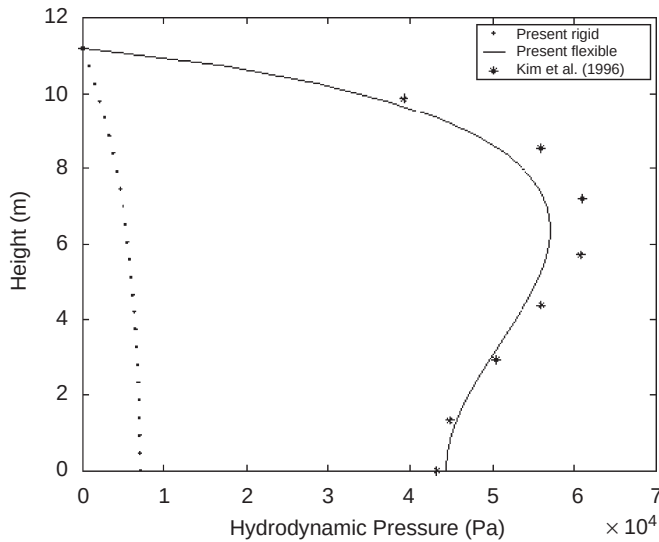


Fig. 6. Hydrodynamic pressure distributions on the wall of the flexible tank at a particular instant due to NS component of El Centro earthquake.

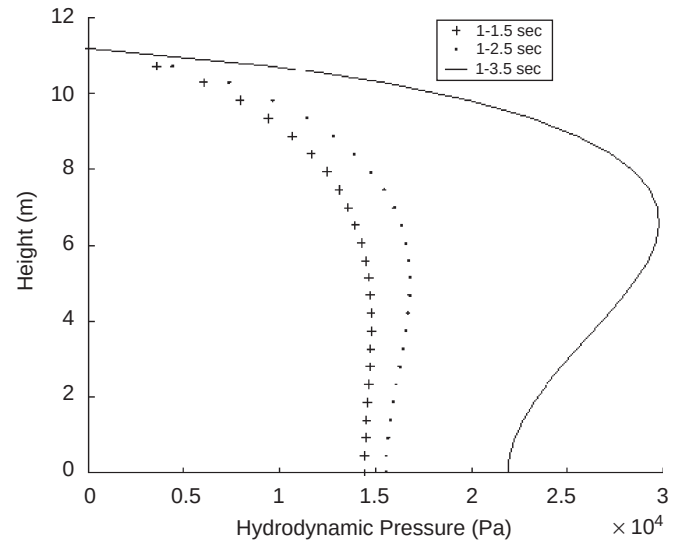


Fig. 7. Hydrodynamic pressure distributions on the wall at certain instants due to NS component of El Centro earthquake.

move in the opposite direction, one inward and the other outward. This mode is termed as the antisymmetric mode. Since the water free surface and the container structure both participate in the oscillation, the oscillating modes may be called coupled fluid–structure mode. In both cases, the shape of the liquid free surface consists of sinusoidal displacements of high wave number. This suggests that the higher sloshing modes are excited during coupled oscillations. The magnitude of free surface displacement is about an order larger than the structural displacement. The structural displacement is shown highly magnified in the figures for clarity.

The natural frequency for the first coupled fluid–structure oscillation mode, which happens to be the symmetric mode, for this tank system is computed to be 20.94 rad/s, while the dry natural frequency is 30.0 rad/s. The frequency becomes 21.18 rad/s if compressibility is neglected. The corresponding frequency for the antisymmetric mode is 21.31 rad/s with compressible water and 21.64 rad/s with incompressible water.

To carry out dynamic studies of the same tank system, the NS component of the El Centro earthquake ground acceleration is applied as the base excitation acting normal to the long sidewall of the tank configuration mentioned above, i.e., along the 56 m long wall. Hence, the 19.6 m × 12.3 m cross section is used for the present analysis. Fig. 6 shows the hydrodynamic pressure at a particular instant on the wall of a rectangular tank. The result compare satisfactorily with the 2D coupled BEM-FEM solution reported in Kim et al. [16]. The pressure on the elastic walls is much higher than on the equivalent rigid walls and the distribution is of quite different nature. The maximum hydrodynamic pressure no longer occurs at the base of the wall as in the rigid case, but shifts upwards to a distance near about $2H/3$ above the base. The upward shifting of the peak hydrodynamic pressure along with the large overall increase in the pressure implies that the bending moment exerted by the hydrodynamic loading is much larger than that predicted by the rigid wall analysis. The differences between the present computation and the computation due to Kim et al. [16] may be attributed to the slightly different time considered and the effects of compressibility and free surface sloshing motion. While Kim et al. [16] have given the pressure corresponding to the peak base shear; in the present solution the instant corresponding to peak wave amplitude at the left free surface node is considered. In addition, Kim et al. [16] did not consider the compressibility of the water and did not fully take in to account the

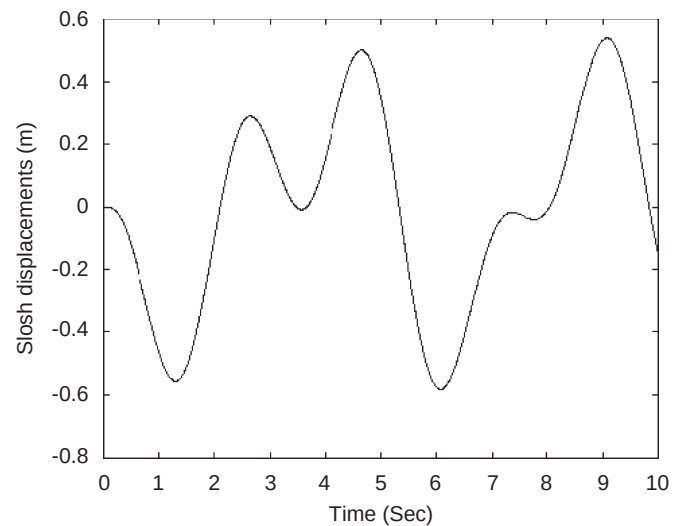


Fig. 8. Time history of free surface displacement at the left wall in a 19.6 m × 12.3 m flexible walled tank with 11.2 m water depth (wall thickness 1.2 m).

sloshing motion. The time evolution of the wall pressure is shown in Fig. 7. It appears that the characteristic distribution takes some time to develop. Fig. 8 shows the free surface displacement history for the case considered. The free surface displacement is almost identical to the rigid case. Since, the wall thickness is quite large, the effect of flexibility on the sloshing motion is virtually negligible. The contributions from the higher modes are marginal. However, as observed from Fig. 6, the effect on the hydrodynamic pressure is quite considerable. This suggests that the inertial response, i.e., the liquid response due to structural acceleration, is much more dominant than the sloshing response.

Fig. 9 shows the hydrodynamic pressure distribution along the wall of a tank of capacity 50 m × 20 m × 10 m with water filled up to a height of 9 m. The tank wall thickness is taken to be 1 m. The tank material has same modulus of elasticity, namely 2.0776×10^{10} Pa, but a density of 2400 kg/m³. A sinusoidal base acceleration of amplitude 1.0 m/s² and frequency of 0.4424 Hz applied normal to the long sidewall is taken as the input motion. The exciting frequency in

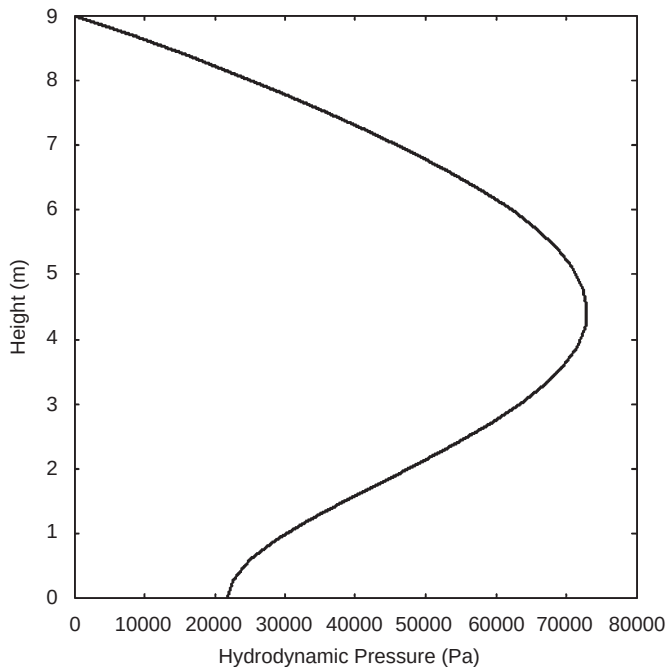


Fig. 9. Hydrodynamic pressure distribution on the wall of the 20m $\times$ 10m flexible tank at a particular instant (1.2s) due to sinusoidal base excitation.

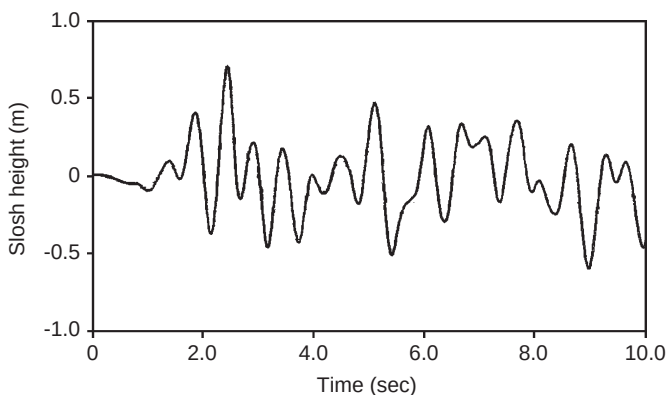


Fig. 10. Free surface displacements at the left wall in the 20m $\times$ 10m flexible walled tank with water depth of 9.0m due to the NS component of El Centro earthquake (wall thickness 0.45 m).

this case is far apart from the coupled natural slosh frequency. Nevertheless, the hydrodynamic pressure distribution demonstrates the characteristic coupled behavior. At the earlier instant of response, the peak is found to be at a little lower distance from the base. In spite of the nearly identical tank configuration, the peak hydrodynamic pressure in this case is significantly higher than the earlier case. This may be due to the difference in relative masses of the fluid and the structure that participate in the motion. The peak pressure seems to be higher if the mass of the structure is reduced compared to the mass of the liquid, other factors remaining more or less unaltered. This also indicates the importance of the inertial response of the hydrodynamic pressure on the wall.

The time history of the free surface elevation at the left wall of the 20m $\times$ 10m tank with liquid depth of 9.0m due to the NS component of 1940 El Centro earthquake is shown in Fig. 10. The tank walls, in this case, are 0.45 m thick. The effect of wall flexibility becomes dominant as the thickness is reduced or the flexibility is increased

and the sloshing motion amplifies. Koh et al. [17] also made similar observations. The sloshing displacements are relatively larger and the presence of the higher modes in the response is clearly evident. It seems that the effect of wall flexibility on the slosh wave amplitude depends on the interaction modes. The effect will be small if the interaction modes due to wall flexibility are away from the pure sloshing modes.

5. Conclusion

A pressure-based Galerkin finite element code that can handle liquid sloshing in a rectangular container has been developed and is coupled with a structural dynamics code. The analysis is restricted to linear problems in the sense that only small amplitude waves (relative to the liquid depth) have been assumed. The pressure formulation has certain advantages in the computational aspect compared to the velocity potential and the displacement-based formulations, as the number of unknown per node is only one. Also, the pressure at the structure–fluid interface is directly obtained which is a significant computational advantage for a coupled simulation. The time integration is performed using either a sequential approach or a fully coupled approach. In the sequential approach the coupling effects are accommodated through iterations. The sequential approach needs much less storage and time. The method has been applied to a number of problems and some typical results are presented that assess the accuracy and applicability of the method. The coupling phenomena are found to have great significance in the case of fluid–structure interaction analysis. The hydrodynamic pressure tends to be amplified and its distribution differs from that of the corresponding rigid container. The pressure distribution develops a much higher bending moment on the container walls. The sloshing motion also amplifies with the increase in wall flexibility.

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